

Last-Margin Gating Under a True Temporal Endpoint Data-Quality Shock

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Abstract

Checkpoint policies are usually evaluated as if the data distribution were fixed throughout training. We instead study a controlled temporal endpoint shock in synthetic sequence classification: labels remain clean until a predetermined subset is flipped only in the final quarter. A last-margin alarm compares the final calibration loss with its running minimum, fires above a threshold frozen on separate FIT seeds, and rolls back from the default last checkpoint to best validation. The alarm detects every shocked hold-out run and no healthy run; relative to last, rollback lowers outer loss by 0.3395. Thus the flag has measurable value against the default endpoint deployment. The boundary is equally important: every shocked trajectory attains its minimum immediately before the shock, so gate, best-val, and early-stop select identical outcomes. E1 therefore demonstrates contamination detection and rollback, not a checkpoint unavailable to best-val. Healthy $m \equiv 0$ is a genuine monotone signature rather than a collapsed readout, but deployment false alarms remain unmeasured on a realistic carrier.

1 Introduction

A checkpoint decision can be undermined by an event that occurs near the end of training. The distinction between a late temporal event and a static corrupted dataset matters: if corruption is active from the first epoch, a poor final checkpoint does not establish that a previously clean candidate was later invalidated. We construct the temporal case directly: a clean prefix ends immediately before a final-quarter label shock. Across the shocked hold-out arm, the checkpoint later selected by best validation is recorded immediately before the shock begins; Figure 1 shows the exact timeline.

We audit a simple deployment signal, the last margin $m = L_{11} - \min_{0 \leq t \leq 11} L_t$, where L_t is calibration loss after epoch t . The signal is read only at the end of the run. If m exceeds a threshold calibrated without hold-out access, the gate raises a contamination flag and deploys the best-validation checkpoint rather than last. Figure 1 shows actual healthy and shocked trajectories, not a schematic.

The experiment is pre-registered and deliberately narrow. Separate FIT seeds calibrate the threshold, and disjoint hold-out seeds settle detection and checkpoint outcomes; Tables 1 and 8 record the complete split. The result changes the interpretation of the alarm. Table 2 shows that the endpoint default is damaged while pre-shock selectors survive, and Table 3 shows that the gate rolls every affected endpoint back. This is measurable flag value. It is not evidence that the gate discovers a checkpoint that best-val misses: best-val, early-stop, and gate agree run by run because the minimum is already present before the late shock.

Our contributions are:

1. A code-level temporal endpoint-shock protocol. Labels are clean through a fixed prefix and flipped for a predetermined affected subset only in the final quarter. The frozen calibration and run inventory are reported in Tables 1 and 8; the paired and operating settlements are reported in Tables 3 and 4.

2. A measured separation between endpoint deployment and alarmed rollback. The flag detects every contaminated run and turns a materially damaged last deployment into the pre-shock best-val outcome, with paired uncertainty reported from the frozen settlement.
3. An explicit boundary on checkpoint novelty and false-alarm claims. Gate, best-val, and early-stop are outcome-identical in this arm, and the healthy false-alarm result comes from a strictly decreasing carrier. We verify that this monotonicity is not whole-cell readout collapse, then reserve realistic false-alarm measurement and shocks that suppress the pre-shock argmin for future experiments.

2 Related work

Weight averaging and checkpoint ensembling provide alternatives to endpoint deployment. Stochastic weight averaging broadens the solution region (Izmailov et al., 2018); exponential moving averages are widely used to stabilize targets and model states (Tarvainen & Valpola, 2017); and model soups combine multiple fine-tuned checkpoints without additional inference cost (Wortsman et al., 2022). We include SWA, EMA, and soup as registered baselines, but our question is different: can a scalar read from the completed trajectory attest that the final endpoint was exposed to a late event?

Robustness work shows that nominal accuracy and robustness objectives can diverge (Tsipras et al., 2019). We use that citation only to motivate corruption as a stress test; it is not evidence about temporal label shocks, nor do we propose a label-noise correction method. Our contribution is the timing-controlled audit and its detection-versus-selection interpretation.

3 Method

3.1 A true temporal endpoint shock

A run produces one calibration loss L_t and one model snapshot after each of $E = 12$ epochs. The shocked mode starts from the same clean training examples as the healthy mode. A predetermined subset of 150 of the 600 training examples is designated as affected. Crucially, designation does not change any label during epochs $0, \dots, 8$. The command-line control `--shock-time-fraction=0.25` sets

$$t_{\text{shock}} = \lceil (1 - 0.25)E \rceil = 9. \quad (1)$$

During epochs 9, 10, 11 only, the affected examples use the deterministic target map $y \mapsto (y + 1) \bmod K$; all other labels remain clean. Epoch shuffling changes batch order but not this onset. The intervention is therefore a time-gated late-training event, rather than a claim that sample identity encodes training time.

The observed timeline validates the intended mechanism. All 64 shocked trajectories attain their minimum at epoch 8, immediately before $t_{\text{shock}} = 9$, and all subsequent shocked points remain above that minimum. Thus the clean candidate that later becomes best-val exists before the intervention, while last is recorded after three affected epochs.

3.2 Last-margin gate

Define

$$m = L_{E-1} - \min_{0 \leq t < E} L_t \geq 0. \quad (2)$$

The gate is

$$\text{gate}(m, \tau) = \begin{cases} \text{best-validation checkpoint,} & m > \tau, \\ \text{last checkpoint,} & m \leq \tau. \end{cases} \quad (3)$$

Firing is both an alarm and a rollback decision. The gate cannot return a checkpoint outside the pair {last, best-val}; consequently, its scientific value must be stated relative to the endpoint deployment it replaces, not as a new optimizer.

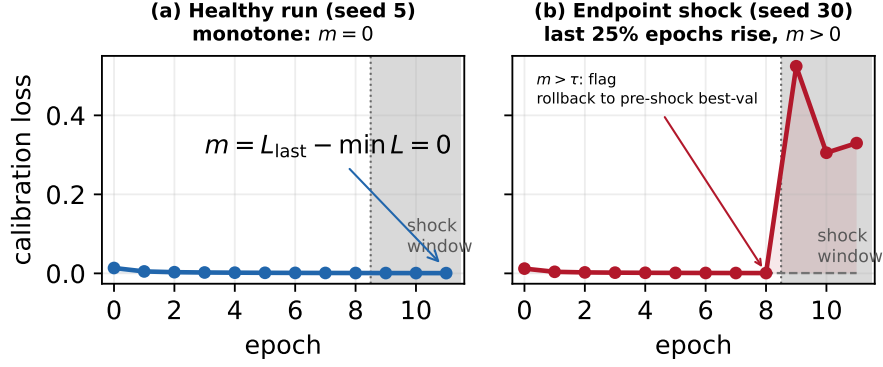


Figure 1: Actual E1 calibration trajectories. Left: healthy seed 5 decreases throughout, so the final point is the minimum and $m = 0$. Right: shocked seed 30 is clean through epoch 8; label flips activate only in the shaded epoch-9–11 window, lifting the trajectory above its pre-shock minimum. The frozen-threshold gate flags the run and returns that pre-shock best-val checkpoint. These representative runs were selected deterministically as closest to the across-run median trajectory; aggregate results use all 64 seeds per arm.

3.3 Frozen calibration and comparison rules

Let H and S be the healthy and shocked last margins from four FIT seeds in each mode. Before examining the hold-out settlement, we freeze

$$\tau = \frac{1}{2} (\max H + \min S) = 0.1354038956. \quad (4)$$

No hold-out trajectory is used to adjust this value. The registered comparison set contains the endpoint, validation, stopping, averaging, soup, and rollback policies summarized with the full configuration in Table 1.

3.4 Settlement estimands

The primary alarm endpoints are recall on shocked runs and false-alarm rate (FAR) on healthy runs, with one-sided 95% Clopper–Pearson bounds. Checkpoint value is measured on the shocked arm by the paired increment

$$\Delta_{\text{inc}} = \mathbb{E} [L_{\text{outer}}(\text{gate}) - L_{\text{outer}}(\text{last})]. \quad (5)$$

The positive control reverses the endpoints, $L_{\text{outer}}(\text{last}) - L_{\text{outer}}(\text{best-val})$, to verify that the late event actually damages endpoint deployment. We use the frozen paired bootstrap intervals and one-sided sign-flip permutation results in the settlement artifact; exact selector identity is reported as an equality rather than converted into a significance claim.

4 Experimental setup

4.1 Carrier and run inventory

The carrier is a self-contained 5-class sequence-classification task trained on CPU. FIT and hold-out seeds are disjoint, and training, calibration, and outer splits are disjoint within each run. Table 1 moves the complete configuration out of prose; Table 8 in the appendix records every settlement unit.

Table 1: Registered E1 configuration. The two fractions refer to different objects: 25% of examples are designated as affected, and their labels change only during the final 25% of epochs.

Component	Setting
Task	5 classes; vocabulary 400; sequence length 12; 6 informative features per class
Splits	train 600; calibration 200; outer 200; disjoint within each run
Model	3-layer transformer; hidden size 64; 4 heads; feed-forward size 128
Optimization	12 epochs; batch 32; learning rate 3×10^{-3} ; weight decay 0.01
Temporal shock	150 affected examples; labels clean in epochs 0–8 and flipped in epochs 9–11
Calibration	FIT seeds 101–104 in healthy and shocked modes; $\tau = 0.1354$
Settlement	hold-out seeds 1–64 in healthy and shocked modes
Selectors	last, best-val, early-stop, SWA, EMA, soup, rollback, margin-gate

Table 2: Mean outer outcomes over 64 hold-out seeds per mode. All rows are recomputed from the E1 raw run records. Bold marks the lowest shocked loss and highest shocked accuracy; equality is retained rather than broken by arbitrary ranking.

Rule	Healthy		Shocked	
	Outer loss	Outer accuracy	Outer loss	Outer accuracy
last	0.00063	1.00000	0.34046	0.95547
best-val	0.00063	1.00000	0.00092	1.00000
early-stop	0.00063	1.00000	0.00092	1.00000
SWA	0.00088	1.00000	0.09780	0.99227
EMA	0.00238	1.00000	0.03900	0.99531
soup	0.00164	1.00000	0.02812	0.99680
rollback	0.00070	1.00000	0.00106	1.00000
margin-gate	0.00063	1.00000	0.00092	1.00000

4.2 Pre-registration and frozen decision rule

The E1 pre-registration was frozen while only the first hold-out seed had completed. It declares four pass conditions: the healthy FAR upper bound is at most 0.10; at least one shocked run is detected; the paired confidence interval for gate-minus-last is wholly below zero; and the paired confidence interval for last-minus-best-val is wholly above zero. The settlement is a direct execution of those conditions. We report all four, including the positive control that could have invalidated the temporal-shock premise.

5 Results

5.1 The endpoint shock damages last, while pre-shock selectors survive

Table 2 reports every registered checkpoint baseline on both modes and both outer metrics. Under the endpoint shock, last has much higher loss than best-val and the gate; its accuracy also falls while the pre-shock selectors remain perfect. Averaging baselines reduce but do not eliminate the loss increase. The table retains every exact value and tied optimum.

The gate, best-val, and early-stop outcomes are exactly equal in all 64 shocked records, for both outer loss and accuracy. Their common checkpoint is the epoch-8 snapshot. The alarm therefore restores an existing pre-shock candidate rather than creating a better checkpoint.

5.2 Paired endpoint effect and rollback increment

Table 3 gives the two frozen paired contrasts. The positive control establishes the damage: last is worse than best-val by +0.3395. Turning on the gate relative to accepting last changes the same quantity with the opposite sign, −0.3395. Both intervals exclude zero

Table 3: Frozen E1 paired outer-loss settlement on the 64 shocked hold-out runs. The two bootstrap intervals were computed separately, so their endpoints are reported exactly as stored. Permutation $p \approx 0$ means zero counter-direction draws in the recomputed one-sided sign-flip settlement.

Contrast	Mean	95% CI	One-sided p
gate – last	−0.3395	[−0.3643, −0.3189]	≈ 0
last – best-val	+0.3395	[0.3187, 0.3634]	≈ 0
gate – best-val	0	exact equality in 64/64	not applicable
gate – early-stop	0	exact equality in 64/64	not applicable

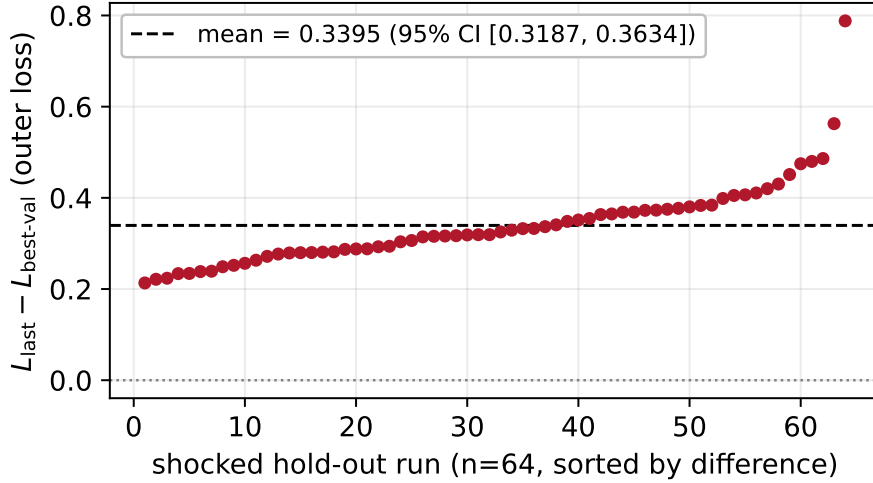


Figure 2: Per-run positive control $L_{\text{outer}}(\text{last}) - L_{\text{outer}}(\text{best-val})$ on 64 shocked hold-out runs, sorted. The mean is +0.3395 and the 95% CI is [0.3187, 0.3634], read directly from the E1 settlement by the plotting script. Every observed difference is positive.

by a wide margin, and the one-sided permutation estimates are approximately zero. The per-run positive controls are shown in Figure 2.

5.3 Alarm operating characteristics

At the FIT-frozen $\tau = 0.1354$, every shocked margin exceeds the threshold and every healthy margin equals zero. Table 4 reports both point estimates and the pre-registered one-sided uncertainty bounds; Figure 3 shows the complete empirical distributions.

5.4 The zero false-alarm rate is a design exclusion, not deployment evidence

$\text{FAR} = 0$ is a real observation on this carrier, but the carrier excludes non-zero healthy margins. Every healthy trajectory is strictly decreasing at every adjacent transition, so its final point is necessarily the minimum and $m \equiv 0$. Table 5 distinguishes this mechanism from whole-cell readout collapse using the complete transition count and independent outer-loss and accuracy readouts.

The exact FAR bound therefore applies only to the registered monotone healthy population. A real fine-tuning carrier with normal stochastic non-monotonicity is required to measure the deployment false-alarm distribution; E1 does not supply that carrier.

Table 4: Alarm operating characteristics at frozen $\tau = 0.1354$. Bounds are one-sided exact 95% Clopper–Pearson intervals in the conservative direction.

Endpoint	Count	Estimate	Exact bound
Recall on shocked runs	64/64	1.0	lower 0.9543
FAR on healthy runs	0/64	0.0	upper 0.0457

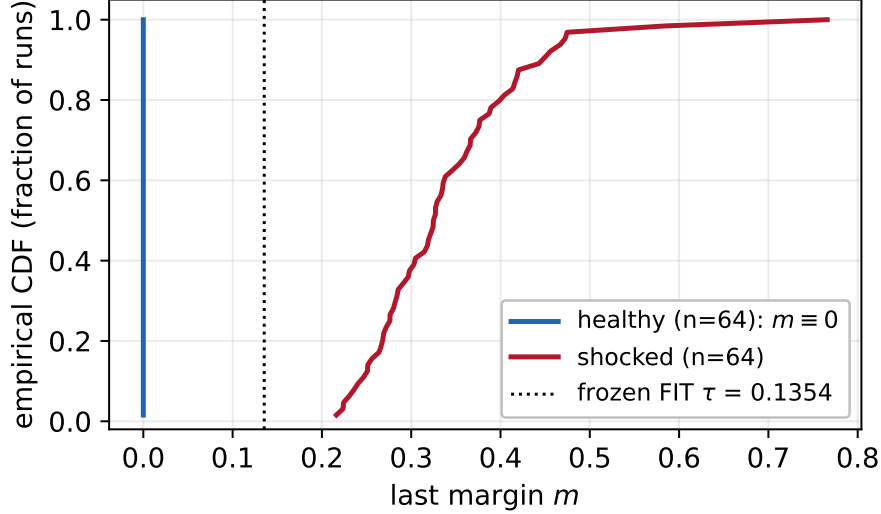


Figure 3: Empirical CDF of the last margin over all E1 hold-out runs. Healthy margins are exactly zero; shocked margins are dispersed but all exceed the FIT-frozen $\tau = 0.1354$. The separation supports the reported E1 operating point only; it is not a deployment calibration curve.

5.5 Interpretation and boundary of the result

5.5.1 The flag has measurable endpoint value, but does not discover a new checkpoint

Relative to the experimental deployment default, the flag changes an observable decision: it identifies every affected run and prevents deployment of the damaged final snapshot. That is the increment quantified by $\Delta_{\text{inc}} = -0.3395$. In contrast, best-val and early-stop need no alarm to obtain the same epoch-8 checkpoint, and their outcomes equal the gate’s in every shocked run. The strongest supported claim is therefore detection plus rollback relative to last, not gate dominance over checkpoint selection.

The decisive next construction is E2: activate a shock early enough, or strongly enough, to suppress or displace the clean argmin so that best-val itself fails. If gate and best-val remain coextensive there, the checkpoint-novelty claim is falsified; if they separate under a frozen rule, that experiment would test a contribution E1 cannot establish.

5.5.2 Distinct overtraining diagnostic

We retain a prior Route-3 diagnostic because it tests a different failure mode, not because it validates E1. The sets defined by positive last margin and low outer accuracy coincide on that arm. Table 6 and Figure 4 report the exact in-sample identity. Thresholds on the last margin and final loss level tie; thus Route 3 supplies no evidence that subtracting the running minimum adds detection information.

Table 5: Independent audit of the healthy $m \equiv 0$ cell. The observations distinguish a genuine monotone signature from an all-zero or non-responsive readout.

Diagnostic	Observed	Interpretation
Strict adjacent decreases	704/704	trajectory changes at every step
Argmin at final epoch	64/64	monotone mechanism for $m = 0$
Best-val outer loss	mean 0.000625	non-zero independent loss readout
Best-val outer accuracy	1.0 in 64/64	task solved, not a dead cell

Table 6: Route-3 overtraining diagnostic, retained unchanged in interpretation and kept separate from E1. Operating points are evaluated in-sample on the same 48 runs; rates are descriptive, not deployment estimates.

Statistic	Recall	FAR
last margin	1.0000	0.0000
final loss level	1.0000	0.0000
first-rise early-stop	0.8929	0.1000
non-monotonicity indicator	1.0000	0.1500

6 Discussion

E1 repairs a timing ambiguity that static corruption cannot resolve. The affected labels are inactive throughout the clean prefix and turn on only in the final quarter. Figure 1 shows the intended causal ordering: a pre-shock minimum, shock onset immediately afterward, and a damaged final deployment. The alarm’s paired benefit and positive control are two signs of the same operational fact, with independently bootstrapped interval endpoints in Table 3.

The experiment also clarifies which policy comparison is meaningful. Against last, the gate has large measured value. Against best-val or early-stop, it has no checkpoint advantage on this arm. SWA, EMA, soup, and rollback remain important baselines because Table 2 shows that several pre-existing policies absorb much of the late shock without an explicit flag. The alarm can still provide attribution, but E1 does not show that attribution is necessary when a pipeline already deploys a robust pre-shock selector.

Table 7 makes the evidential boundary explicit. In particular, the healthy result is internally valid but externally narrow: strict monotonicity is verified as a real signal, yet it removes the variation needed for a realistic FAR test.

Table 7: Claim boundaries and decisive next evidence. A checked E1 observation is not promoted beyond the population or policy comparison that generated it.

Question	What E1 establishes	What remains open
Temporal integrity	Flips activate only in epochs 9–11	Other shock types and durations
Alarm detection	Recall 1.0 at frozen τ	Recall on a realistic carrier
Endpoint value	Gate beats last by 0.3395 loss	Gate advantage over best-val (E2)
False alarms	FAR 0.0 on verified monotone runs	FAR with natural non-monotonicity (E6)
Route-1 probe	Prior 36-run probe used a smaller, different carrier	No transport of its rates to E1

7 Limitations

1. Synthetic scale and intervention. The carrier is a small CPU-trained classifier. The intervention flips a predetermined affected subset only in the final three epochs. This is a true late-training event, but it is not an external incident occurring after

training has ended, and its effect size must not be transported to large-model fine-tuning.

2. Healthy variation is excluded. Table 5 verifies that healthy $m \equiv 0$ is a genuine monotone signature rather than readout collapse. Nevertheless, the absence of healthy non-monotonicity makes FAR = 0 partly a design exclusion. E6 requires a realistic carrier before any deployment-safety claim.
3. Best-val is not defeated. Every shocked argmin occurs before shock onset, as shown in Figure 1, and gate, best-val, and early-stop therefore agree exactly. E1 measures the value of a flag relative to last; E2 must cover or suppress the argmin to test whether a gate can add checkpoint value beyond best-val.
4. Threshold transport is untested. The midpoint $\tau = 0.1354$ is blind to hold-out but calibrated on the same synthetic mechanism. No E1 threshold sweep is used to claim a safe band, and the frozen value is not a recommended deployment threshold.
5. Secondary diagnostics use distinct carriers. Route 3 is an in-sample overtraining diagnostic, and the prior Route-1 probe uses a downsized configuration. Neither supplies an E1 FAR estimate or a substitute for a realistic healthy carrier.

8 Conclusion

A time-gated endpoint shock in the final quarter of training creates a clean before-and-after checkpoint audit. At the frozen threshold, the last-margin gate detects every shocked run, produces no alarms on verified-monotone healthy runs, and materially improves outer loss relative to last (Tables 3 and 4). The equally important boundary is exact selector identity: every shocked argmin lies before the shock, so gate, best-val, and early-stop return the same outcomes. E1 therefore supports a measurable contamination flag and rollback against endpoint deployment, not a new checkpoint beyond best-val. Testing that stronger claim and measuring realistic false alarms require E2 and E6, respectively.

Reproducibility

The frozen pre-registration, FIT threshold, 128 raw hold-out records, settlement JSON, verdict, and deterministic figure script are included in the anonymous workspace. Table 8 enumerates the run budget. Quantitative figures are generated only from registered E1 records; the conceptual image-generation candidate described in the supplementary disclosure is not used as experimental evidence. AI assisted language editing; every claim and number was checked against the registered artifacts.

References

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A Run inventory and provenance

Table 8: E1 run inventory. FIT runs only calibrate τ ; all point estimates and uncertainty reported for E1 use the 128 hold-out runs.

Phase	Seeds	Modes	Runs
FIT calibration	4	healthy, shocked	8
Hold-out settlement	64	healthy	64
Hold-out settlement	64	shocked	64
Total	136 (8 FIT + 128 hold-out)		

The temporal mechanism is implemented in `experiments/code/combined.py`; the frozen run and settlement procedure is in `experiments/code/run_e1_temporal_shock.sh`; the pre-registration is `experiments/prereg/PREREG_E1_TEMPORAL.md`; and the frozen numerical settlement is `results/three_arm_temporal/E1_SETTLEMENT_frac0.25.json`. The representative and aggregate figures are reproduced by `experiments/code/p5_e1_figs.py`.

B Distinct Route-3 diagnostic

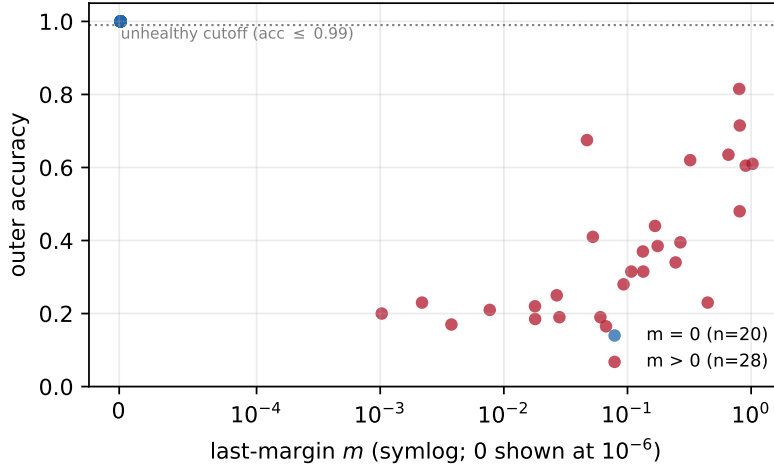


Figure 4: Route-3 overtraining scatter, retained as a distinct diagnostic. On these 48 runs, $m > 0$ and outer accuracy ≤ 0.99 identify the same 28 runs. The identity explains the perfect in-sample last-margin row in Table 6; it does not provide an E1 operating estimate.

C Statistical details

Recall and FAR bounds are one-sided 95% Clopper–Pearson bounds. The two paired 95% confidence intervals in Table 3 use the bootstrap distributions frozen in the E1 settlement. The sign-flip permutation results were recomputed with 20,000 one-sided draws and correct elementwise signs; the stored empirical values are zero, which we report as $p \approx 0$ rather than as mathematical probability zero. Exact equality statements for gate, best-val, and early-stop come from recordwise equality of both stored outer loss and stored outer accuracy, not a null test.